*Science in China Series F: Information Sciences*
© 2009 SCIENCE IN CHINA PRESS
Springer

www.scichina.com
info.scichina.com
www.springerlink.com

# Accurate evaluation of Green's functions in a layered medium by SDP-FLAM

SONG Zhe†, ZHOU HouXing, HU Jun & HONG Wei

State Key Lab of Millimeter Waves, School of Information Technology and Engineering, Southeast University, Nanjing 210096, China

**Based on local Taylor expansions on the complex plane, a method for fast locating all modes (FLAM) of spectral-domain Green's Functions in a planar layered medium is developed in this paper. SDP-FLAM, a combination of FLAM with the steepest descent path algorithm (SDP), is employed to accurately evaluate the spatial-domain Green's functions in a layered medium. According to the theory of complex analysis, the relationship among the poles, branch points and Riemann sheets is also analyzed rigorously. To inverse the Green's functions from spectral to spatial domain, SDP-FLAM method and discrete complex image method (DCIM) are applied to the non-near field region and the near filed region, respectively. The significant advantage of SDP-FLAM lies in its capability of calculating Green's functions in a layered medium of moderate thickness with loss or without loss. Some numerical examples are presented to validate SDP-FLAM method.**

Green's functions, layered medium, fast all modes method, surface wave poles, leaky wave poles, branch cut, Riemann sheet

With the development in microwave and millimeter wave integrated circuit design and VLSI technology, multi-functionality, high-performance and sub-miniaturization will be the advancing trends in the near future. Meanwhile, with the progress in microwave monolithic integrated circuit (MMIC) and low temperature co-fired ceramic (LTCC) techniques and with the popularity in the conceptions of system on chip (SoC) and system in package (SiP), more and more attention has been paid to the rigorous, accurate and fast modeling and simulation method of layered circuits.

The spatial-domain Green's functions in a layered medium are vital parts in full wave analyses of complicated microstrip structures using the method of moment (MoM). Based on the geometry features of a layered medium, the spectral-domain Green's functions can be expressed in closed forms, and hence can be inversed to spatial domain through the well-known Sommerfeld integrals (SI)[1], whose integrands are Bessel/Hankel functions. Both pole point and branch point singularities of the spectral-domain Green's functions as well as the high oscillation of Bessel/Hankel functions result in slow attenuation and strong oscillation of the Sommerfeld integrals on their paths so that the direct calculations of the Sommerfeld integrals are quite time-consuming, which is a bottle-

Received February 26, 2008; accepted May 14, 2008; published online March 25, 2009
doi: 10.1007/s11432-009-0021-0
†Corresponding author (email: zhesong@emfield.org)
Supported by the National Natural Science Foundation of China (Grant No. 60621002), and the State Key Development Program for Basic Research of China (Grant No. 2009CB320200)

**Citation:** Song Z, Zhou H X, Hu J, et al. Accurate evaluation of Green's functions in a layered medium by SDP-FLAM. Sci China Ser F-Inf Sci, 2009, 52(5): 867–875, doi: 10.1007/s11432-009-0021-0

neck in modeling and analyzing the planar layered medium structures based on the method of moment.

The steepest descent path (SDP)[2,3] method is the foremost fast algorithm for calculating the Sommerfeld integrals. The main idea of SDP is to select such a deformed integral path on complex plane that the convergence of the integration can be accelerated. However, because of the restriction in a numerical computer, accurate calculation of complex variable Bessel/Hankel functions is still difficult. In 1984, Lindell[4] suggested the accurate complex image method. During 1988 and 1991, Fang et al.[5,6] proposed the discrete complex image method (DCIM), which laid the foundation for semi-numerical and semi-analytical methods for inversing the spectral-domain Green functions to spatial domain. DCIM[7,8] decomposes the spectral-domain Green's functions into three parts, namely quasi-static term, surface wave term, and complex image term, among which the last is obtained with the function fitting technique. In 1995, Sakar, et al.[9] used the generalized pencil of function (GPOF) to do fitting in the complex image part, which is more robust and reliable than the previous Prony[5] method. In 1996, Aksun et al.[10,11] developed the original DCIM into the two-level version, which can enhance the accuracy and adaptability simultaneously without pole extractions. In 1997, Michalsky and Mosig[12] systematically derived the integral equation formulations of Green's functions in a multilayered medium and also provided the mixed-potential integral equation (MPIE). In 1999, Cui and Chew[13] obtained the further acceleration of SDP method based on the symmetrical characteristic of the Green's functions. From 2000 up to now, the use of DCIM to evaluate the spatial-domain Green's functions in MPIE becomes more and more popular for full wave analysis of multilayered medium structures based on MoM. In this aspect, Ling and Jin et al.[14−17], Li and Liu et al.[18−20], Boix, Medina and Horno et al.[21], Simsek and Liu et al.[22,23] made their contributions. However, most researches have been focusing on thin layer structures. Beside the works mentioned above, there are also some methods from different points of view. In 1998, Hsieh and Kuo[24] introduced the fast Hankel transform (FHT) method into this area. In 1999, Song and Hong et al.[25] used the Chebyshev pseudo spectrum expansion combined with Krylov subspace order-reduction techniques to calculate spatial-domain Green's functions. In 2006, Kourkoulos and Cangellaris[26] proposed an accurate approximation of spectral domain Green's functions as a set of spherical and cylindrical waves based on vector fitting (VECTFIT). Meanwhile, many researchers, including King and Sandler et al.[27,28], Zhang et al.[29], Li et al.[30] did a lot of work in this area. The significant advantage of DCIM is that it can get the spatial-domain Green's functions without any numerical calculation. However, DCIM is not suit for a moderately thick layered medium, especially the far field region. Moreover, its precision in the far field region could not be improved even if the number of complex images is further increased. In 2003, Teo et al.[31] presents a method for pole extraction. In 2006, Neve and Paknys[32] suggested using the Newton-Raphson algorithm to find poles. Then, in 2007, Tsang, Ong and Wu et al.[33−35] proposed a fast all modes (FAM) method combined with numerical modified steepest path (NMSP) to calculate Sommerfeld integrals. This method could deal with layered media with a moderate thickness, and its significant advantage is that it can find all modes without any contour integration in a proper range with the origin as the center on the complex plane, namely surface wave modes, leaky wave modes and improper modes. However, this method could not ensure the validity in a large range.

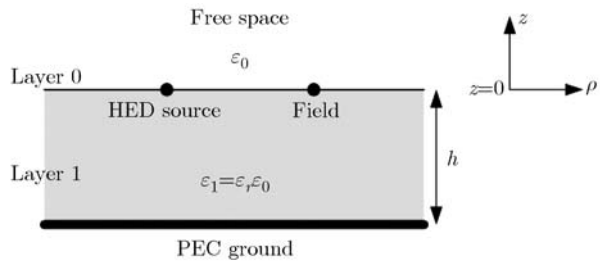


**Figure 1** Model of single layered medium with source (HED) and field points locate on boundary.

In this paper, based on local Taylor expansions on the complex plane, a method for fast locating

all modes (FLAM) of spectral-domain Green's functions in a planar layered medium is developed. SDP-FLAM, a combination, of FLAM with the steepest descent path algorithm (SDP), is applied to the accurate evaluation of the spatial-domain Green's functions in a layered medium. According to the theory of complex analysis, the relationship among the poles, branch points and Riemann sheets is also analyzed rigorously. To inverse the Green's functions from spectral to spatial domain, SDP-FLAM method and discrete complex image method (DCIM) are applied to the non-near field region and the near filed region, respectively. The significant advantage of SDP-FLAM lies in its ability to calculate Green's functions in a layered medium of moderate thickness with loss or without loss. Some numerical examples are presented to validate SDP-FLAM method. The model of all the numerical examples is illustrated in Figure 1, where Layer 0 is free space and Layer 1 is medium with $\varepsilon_r$ as the relative permittivity. A PEC boundary lies under Layer 1. Every layer has a permeability $\mu_0$. The source point is a horizontal electric dipole (HED) located at interface between Layer 0 and Layer 1, and the field point has the same height as the source point. Only TE mode is considered and the time factor $e^{j\omega t}$ is adopted in this paper.

## 1 Theory analysis

### 1.1 Spectral-domain Green's function and Sommerfeld integral

The spectral-domain Green's function of the model depicted in Figure 1 could be obtained from transmission line theory[12] as follows:

$$j\omega\mu_0 \cdot \widetilde{G}_A^{xx} = V_i^h, \tag{1}$$

where $\omega$ stands for angular frequency, $\mu_0$ for permeability of free space, and $V_i^h$ for a set of TE type transmission lines excited by a unit of shunt current. According to the theory of transmission line Green's function[12], we have

$$\widetilde{G}_{xx}^A = \frac{1}{\mathrm{j}\omega\mu_0} \cdot \frac{Z_m^h}{2} \cdot \left[\mathrm{e}^{-\mathrm{j}k_{mz}|z-z'|} + M_m^h \cdot \sum_{s=1}^{4} B_{ms}^h \cdot \mathrm{e}^{-\mathrm{j}k_{mz}\gamma_{ms}}\right], \tag{2}$$

where $Z_m^h$ and $k_{mz}$ stand for characteristic impedance and wave vector in $z$ direction of the source layer, respectively. $z$ and $z'$ stand for the coordinates of source point and field point in $z$ direction, respectively, while $M_m^h, B_{ms}^h$ and $\gamma_{ms}$ are coefficients depending on the type (TE) of transmission line. Hence, the spectral-domain Green's function can be expressed as

$$\widetilde{G}_{xx}^A(k_\rho) = \frac{\sin(k_{1z}h)}{k_{1z}\cos(k_{1z}h) + \mathrm{j}k_{0z}\sin(k_{1z}h)}. \tag{3}$$

The spatial-domain Green's function can be obtained through Sommerfeld integrals as follows:

$$G_{xx}^A(\rho, z|z') = \frac{1}{2\pi}\int_0^\infty \widetilde{G}_{xx}^A(k_\rho, z|z') J_0(k_\rho\rho) k_\rho dk_\rho, \tag{4a}$$

$$G_{xx}^A(\rho, z|z') = \frac{1}{4\pi}\int_{-\infty}^\infty \widetilde{G}_{xx}^A(k_\rho, z|z') H_0^{(2)}(k_\rho\rho) k_\rho dk_\rho. \tag{4b}$$

### 1.2 Accurate location of all poles of spectral Green's function

It is easily known from formulas (3) and (4) that the poles are all contained in the denominator of (3), and hence this part can be extracted out for convenience, i.e.

$$f = k_{1z}\cos(k_{1z}h) + \mathrm{j}k_{0z}\sin(k_{1z}h) = 0. \tag{5}$$

Using variable substitution $u = k_{1z}h$ and $v = jk_{0z}h$ in (5), one has[35]

$$f(u, v) = v\sin u + u\cos u = 0, \tag{6}$$

where $u$ and $v$ satisfy

$$u^2 + v^2 = (k_1^2 - k_0^2)h^2 = r^2. \tag{7}$$

Substituting (7) into the square of (6), one can get a quite simple formula

$$\cos 2u + \left(\frac{2u^2}{r^2} - 1\right) = 0, \tag{8}$$

which is a transcendental equation. Substituting a truncated Taylor expansion with truncation index $N_{\mathrm{MAX}}$ into (8), one can obtain an approximate polynomial equation

$$\sum_{n=0}^{N_{\mathrm{MAX}}} \frac{\cos(a + n\pi/2)}{n!}(t)^n + \frac{1}{2r^2}t^2$$

$$+\frac{a}{r^2}t+\left(\frac{a^2}{2r^2}-1\right)=0, \tag{9}$$

where $u=(t+a)/2$ and $a$ is the expansion point with regard to $2u$. A root of (9) can be accepted as that of (8) as long as the truncation error of (9) is small enough. Polynomial equation (9) can be effectively solved by using an available method, such as the Matlab function "roots". Through moving expansion point $a$, one can locate all the roots of (9) on complex $u$ plane, namely all the poles of the spectral-domain Green's function. The distribution of all the roots in (9) is now illustrated in Figure 2(a).

All the poles can be classified into three different kinds of modes according to their physical senses[36,37]:

1) Surface wave poles (SWP), which correspond to surface wave, propagate along the boundary surface;

2) leaky wave poles (LWP), which correspond to leaky wave, attenuate in horizontal direction but exponentially increase in vertical direction;

3) remainder poles, which have no corresponding physical sense.

There should be some rules for moving the expansion points. To explain it concisely, we may suppose that the region of interest is only scanned from left to right, as shown in Figure 2(b). First, we use a Taylor expansion with expansion point $A$ and get a convergent radius $l_A$. Then, the next expansion point $B$ is selected such that $B \leqslant A+l_A$. After obtaining a convergent radius $l_B$, the third expansion point $C$ is selected such that $C \leqslant B+\min(l_A, l_B)$, and so on. In other words, the minimum convergent radius obtained before the current moving can be selected as an upper bounder of the current moving distance. One can also add the vertically moving rules if necessary. The selection of truncation index $N_{\mathrm{MAX}}$ depends on the moving rules of expansion points, but has no effect on the accuracy of the calculated results of roots. Usually, one can set $N_{\mathrm{MAX}}=80$ or $N_{\mathrm{MAX}}=100$, with which the absolute error of (9) lies between $10^{-15}$ and $10^{-13}$. Therefore, the roots calculated by the proposed method did not have to be further iterated by using the Newton-Raphson algorithm. Comparing with the method in ref. [35], the method in this paper can not only save the computing time, but also reduce the arithmetic complexity. It is clearly seen from Figure 2(a) that the distribution of the roots is conjugately symmetric, as can be validated by substituting $u$ and $u*$ into (8).

According to the theory of complex analysis[38], the number $N_{\mathrm{zeros}}$ of zeros and the number $N_{\mathrm{poles}}$ of poles of a meromorphic function $f(z)$ inside a simple closed curve $C$ satisfy the following relationship:

$$\oint_C \frac{f'(z)}{f(z)}dz = 2\pi j(N_{\mathrm{zeros}}-N_{\mathrm{poles}}), \tag{10}$$

where $f'(z)$ is the derivative of $f(z)$. Now, based on this formula, one can examine the number of poles obtained by the proposed method. We select a rectangular closed contour $A-B-C-D-A$ as shown in Figure 2(a), and calculate the contour

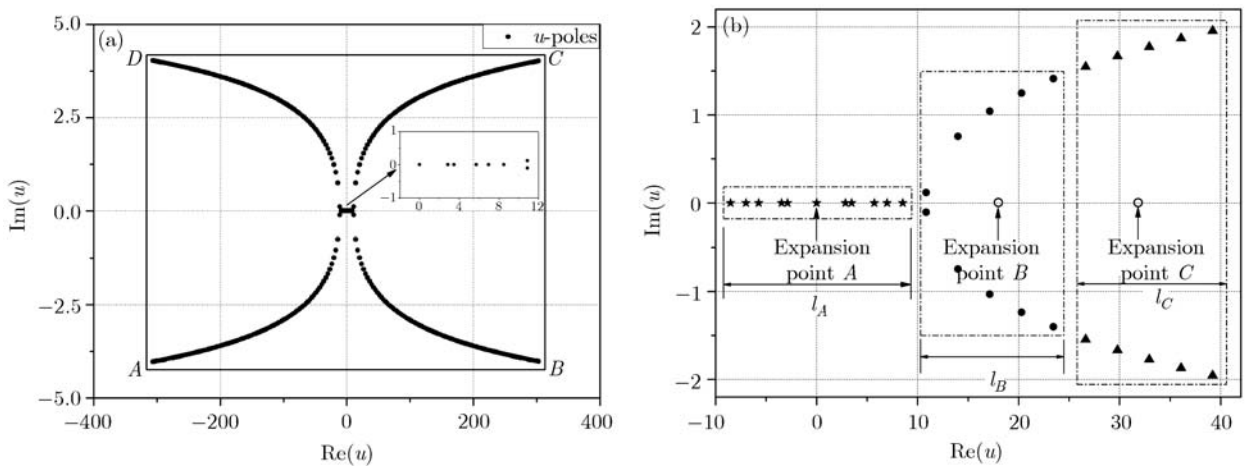


**Figure 2** (a) Distribution of roots in (9) on complex $u$ plane; (b) sketch of moving expansion points.

**Table 1** Comparison of number of poles gained by Taylor expansion and contour integral

| Expansion range of $u$ | Number of roots from Taylor expansion | Number of roots from contour integral |
|---|---|---|
| $[-20, 20]$ | 23 | 23 |
| $[-50, 50]$ | 63 | 63 |
| $[-100, 100]$ | 127 | 127 |
| $[-250, 250]$ | 319 | 319 |
| $[-500, 500]$ | 635 | 635 |

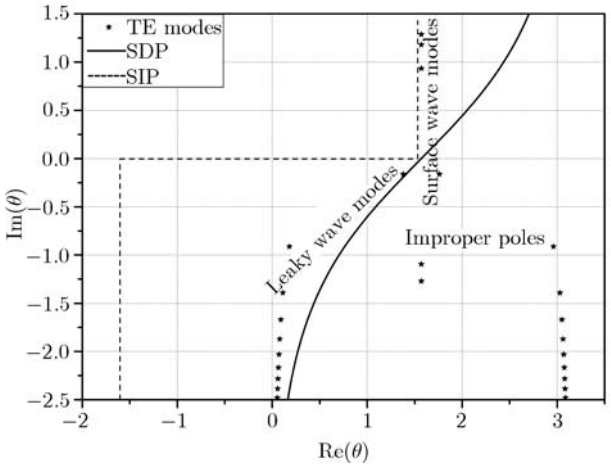


**Figure 3** Locations of SIP, SDP and all TE modes of example 1 on complex $\theta$ plane.

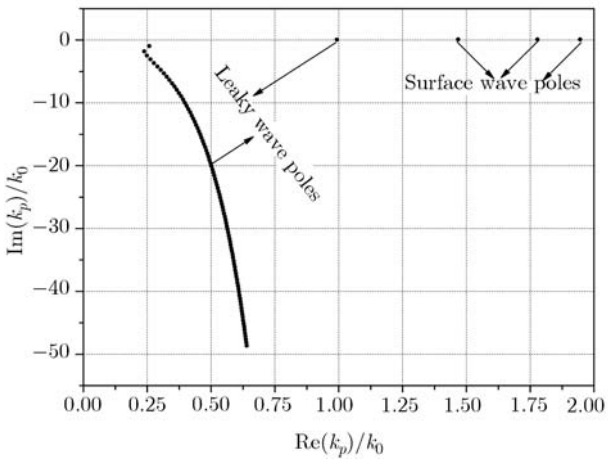


**Figure 4** Locations of SWP, LWP and branch cut of example 1 on $k_\rho$ plane.

integral on it. A comparison between the numbers of poles obtained by the two methods is shown in Table 1. It can be found from Table 1 that the proposed method is applicable in a very wide range.

After the poles on complex $u$ plane are located, they can be mapped onto complex $k_\rho$ plane by means of $k_\rho = \sqrt{k_1^2 - (u/h)^2}$. Since only the SWP and LWP are needed in the calculation of the Sommerfeld integral, the remainder poles should be excluded in advance. According to the relations $k_\rho = k_0 \sin\theta$ and $k_{0z} = k_0 \cos\theta$, the poles on complex $k_\rho$ plane can be again mapped onto complex $\theta$ plane, on which the steepest descent path can be characterized as $\mathrm{Re}(\sin\theta) = 1$. It can be found from Figure 3 that the SWP and LWP distribute between Sommerfeld integral path (SIP) and steepest descent path (SDP), and Figure 4 depicts the distribution of them on complex $k_\rho$ plane.

### 1.3 Topology analysis of the spectral-domain Green's function

With Hankel function as the integrand kernel, the Sommerfeld integral path in (4b) extends from $-\infty$ to $+\infty$[1]. According to the branch cuts of Hankel function and spectral-domain Green's function as well as the locations of SWP and LWP, the topology of spectral-domain Green's function can be determined[7,39], as shown in Figure 5, in which the Riemann sheets were defined by $k_z = \pm\sqrt{k_0^2 - k_\rho^2}$.

The deformed SIP rounds all the singularity on the first Riemann sheet and extends to the infinite along the real axis. SWP, which belongs to the top Riemann sheet, is located between the maximum and minimum wave number of the model if the medium is lossless, and will move into the fourth quadrant when the medium is lossy. LWP is loca-

ted on the bottom Riemann sheet. Obviously, SWP and LWP lie in different Riemann sheets. Hence, Sommerfeld integral can be expressed as a sum of three different parts: 1) residues of all the surface wave poles; 2) residues of all the leaky wave poles; 3) integral contribution of branch cut, that is,

$$G_A = G_A^{\Gamma} - \mathrm{j}2\pi \cdot D \cdot \left( \sum_i^{N_{\mathrm{SWP}}} R_{k\rho i} + \sum_j^{N_{\mathrm{LWP}}} R_{k\rho j} \right), \quad (11)$$

where $G_A^{\Gamma}$ stands for the integral contribution with path shown in Figure 5. $D$ is a coefficient depending on the media and $R_{k\rho}$ stands for the residue. $N_{\mathrm{SWP}}$ and $N_{\mathrm{LWP}}$ stand for the numbers of SWP and LWP, respectively. One should pay more attention to the calculation of residues, since the complex variables should be signed according to different Riemann sheets.

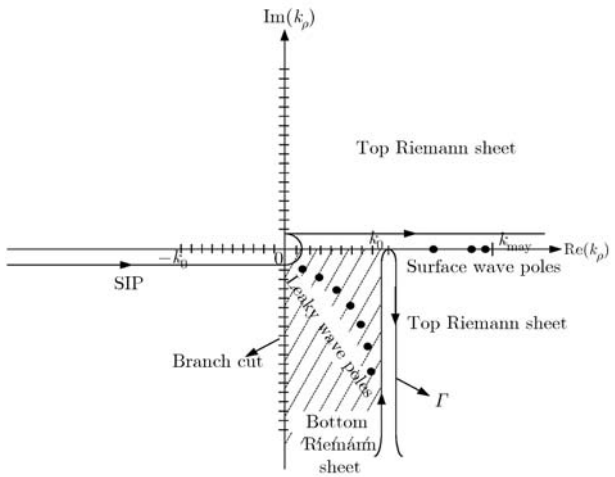


**Figure 5** Topology of spectral Green's function on complex $k_\rho$ plane.

### 1.4 Accurate evaluation of the spatial-domain Green's function

1.4.1 Non-near field region: $\rho > 0.05\lambda$. Firstly, the residues in (11) can be calculated by a set of formulas below[35]:

$$R_{k\rho p} = H_0^{(2)}(k_{\rho p}\rho) \frac{P(k_{\rho p})}{[dQ(k\rho)/dk\rho]|_{k\rho = k\rho p}}, \quad (12a)$$

$$P(k_{\rho p}) = \mathrm{j}2\sin(u) \cdot k_{\rho p} \cdot h, \quad (12b)$$

$$[dQ(k\rho)/dk\rho]|_{k\rho = k\rho p} = \left( -\frac{k_{\rho p} h^2}{u} \right) \cdot \left[ \cos(u) - u\sin(u) - \frac{u}{v}\sin(u) + v\cos(u) \right]. \quad (12c)$$

Based on (6), one can build a one-to-one mapping relation:

$$v = -u \cdot \cot(u). \quad (13)$$

With relation (13), the residues can be easily calculated without selecting signs on different sheets. It is known from the large argument asymptotic expression of Hankel function that the Hankel function in (12) exponentially attenuates with $k_{\rho p}\rho$ growing up. Hence, one can use such an inequality, $|\mathrm{Im}(k_{\rho p})\rho| < N$, to wash out the leaky wave poles.

Secondly, the steps for calculating the integral contribution $G_A^{\Gamma}$ in (11) are as follows:

Step 1. Use the relation $k_\rho = k_0 - jx^2/(2\rho)$ to transform the integral path on complex $k_\rho$ plane to the whole real axis on complex $x$ plane.

Step 2. Determine the integrands and integral paths according to different Riemann sheets:

$$G_A^{\Gamma} = G_A^{\Gamma 1} + G_A^{\Gamma 2}, \quad (14a)$$

$$G_A^{\Gamma 1} = \frac{-\mathrm{j}\mu_0}{8\pi} \int_{-\infty}^{0} \frac{x}{\rho} \cdot \frac{2k_\rho}{\mathrm{j}k_{0z}^{R1} + k_{1z}\cot(k_{1z}h)} H_0^{(2)}(k_\rho \rho) dx, \quad (14b)$$

$$G_A^{\Gamma 2} = \frac{-\mathrm{j}\mu_0}{8\pi} \int_{0}^{+\infty} \frac{x}{\rho} \cdot \frac{2k_\rho}{j} k_{0z}^{R2} + k_{1z}\cot(k_{1z}h) H_0^{(2)}(k_\rho \rho) dx, \quad (14c)$$

$$\begin{cases} k_{0z}^{R1} = \sqrt{k_0^2 - k_\rho^2}, \\ k_{0z}^{R2} = -\sqrt{k_0^2 - k_\rho^2}, \end{cases} \quad (14d)$$

where $D = -\mathrm{j}\mu_0/(8\pi)$. By the way, (14) is calculated by the adaptive Simpson integral[40].

1.4.2 Near field region: $\rho < 0.05\lambda$. The Sommerfeld integral calculated by SDP-FLAM is integral (4b), whose integrand contains Hankel function. The origin is the branch point of Hankel function (also a singular point), from which a branch cut extends to $-\infty$ along the real axis, and SDP-FLAM does not involve the integral contribution of this branch cut, so SDP-FLAM is not applicable to the near field Green's function. Here, we calculate the near field Green's function by one-level discrete complex image method. The details are presented as follows:

Step 1. Rewrite the spectral-domain Green's function in a simple form[5]:

$$\widetilde{G} = A \cdot \frac{F}{2\mathrm{j}k_{0z}}, \tag{15a}$$

where $A$ is a coefficient depending on media.

Step 2. Extract the quasi-static term (an asymptotic term)

$$F_{Q-S} = \lim_{k_\rho \to \infty} \widetilde{G}(k_\rho, z|z'). \tag{15b}$$

Step 3. Extract the surface wave term

$$F_{\mathrm{SWP}} = 2\mathrm{j}k_{0z} \cdot \sum_i \frac{2k_{\rho i} \cdot \mathrm{Res}_i}{k_\rho^2 - k_{\rho i}^2}. \tag{15c}$$

Step 4. Use GPOF algorithm to fit the rest part:

$$F - F_{Q-S} - F_{\mathrm{SWP}} = \sum_{i=1}^{Nc} a_i \mathrm{e}^{-\mathrm{j}k_{0z}b_i}. \tag{15d}$$

Step 5. Inverse the three parts to spatial domain by Sommerfeld Identities, respectively:

$$\frac{\mathrm{e}^{-\mathrm{j}k_0 r}}{4\pi r} = \frac{1}{4\pi} \int_{-\infty}^{+\infty} \frac{\mathrm{e}^{-\mathrm{j}k_{0z}|z|}}{\mathrm{j}2k_{0z}} H_0^{(2)}(k_\rho \rho) k_\rho dk_\rho. \tag{15e}$$

$$\begin{cases} G_A^{xx}(\rho) = \dfrac{A}{4\pi}\left(F_{st} \cdot \dfrac{\mathrm{e}^{-\mathrm{j}k_0\rho}}{\rho} - 2\pi j \sum_i \mathrm{Res}_i \right. \\ \qquad \left. \cdot H_0^{(2)}(k_{\rho i}\rho) k_{\rho i} + \sum_{i=1}^{N_c} a_i \dfrac{\mathrm{e}^{-\mathrm{j}k_0 r_i}}{r_i}\right), \\ r_i^2 = \rho^2 + b_i^2. \end{cases} \tag{15f}$$

## 2 Numerical examples

To validate the correctness and efficiency, we provide four numerical examples here corresponding to different thicknesses, loss cases and frequencies. We used Matlab to do programming and $|\mathrm{Im}(k_{\rho p})\rho| < 15$ as a rule for washing out the leaky wave poles. All the positions of surface wave poles are listed in Table 2. To save page space, all the results are accuracy cut off at $10^{-4}$ though they can actually reach up to $10^{-15}$. The last column in Table 2 shows the total CPU time cost for calculation. The spatial-domain Green's functions are shown in Figures 6—9. It can be clearly seen from the obtained data that all the SWP and LWP of the model in Figure 1 can be accurately located

**Table 2** Model parameters, locations of SWP and computation time

| No. | Frequency (GHz) | $h$ | $\varepsilon_r$ | Positions of SWP($k_0$) | CPU time (s) |
|---|---|---|---|---|---|
| 1 | 8 | $1\lambda$ | 4 | 1.4694, 1.7801, 1.9470 | 1.547 |
| 2 | 8 | $2\lambda$ | 4 | 1.1308, 1.4051, 1.6076, 1.7580, 1.8674, 1.9421, 1.9857 | 1.438 |
| 3 | 8 | $1\lambda$ | 4–0.1j | 1.4697–0.0313j, 1.7803–0.0273j, 1.9471–0.0255j | 3.438 |
| 4 | 12 | $1\lambda$ | 4 | 1.4694, 1.7801, 1.9470 | 1.578 |

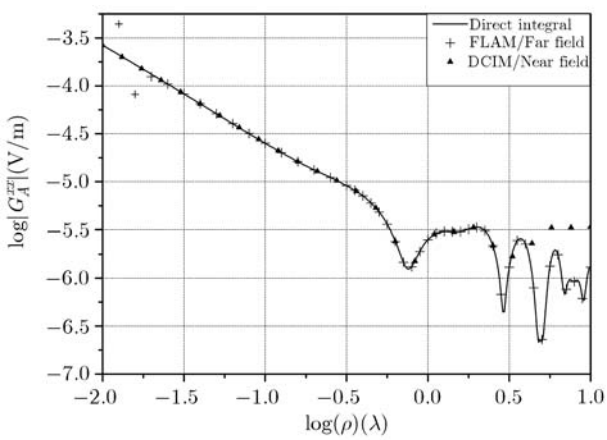


**Figure 6** Spatial Green's function in Example 1.

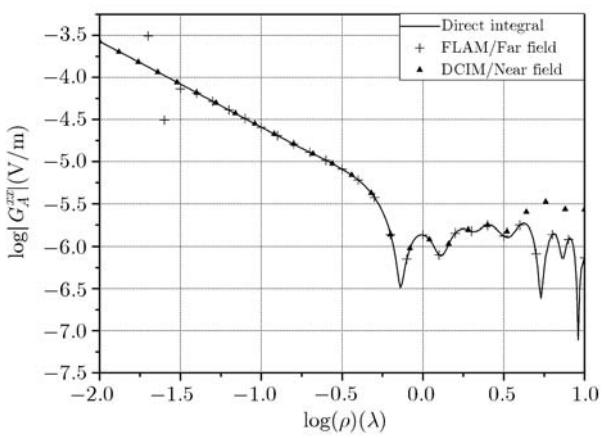


**Figure 7** Spatial Green's function in Example 2.

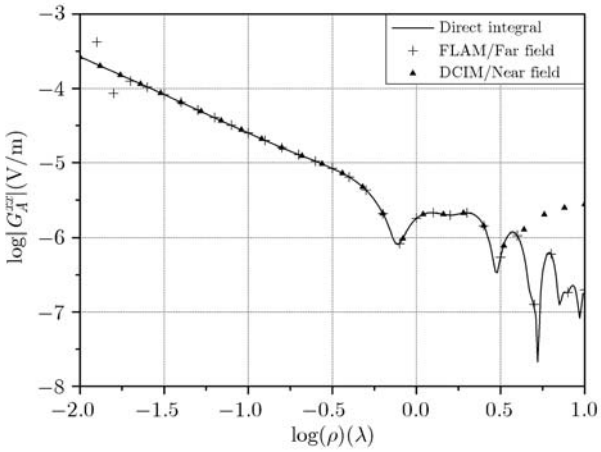


**Figure 8** Spatial Green's function in Example 3.

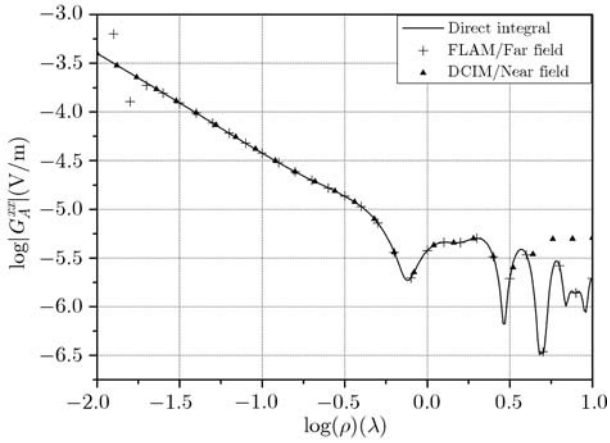


**Figure 9** Spatial Green's function in Example 4.

by the proposed method, and the spatial-domain Green's functions provided by SDP-FLAM combined with DCIM agree well with the accurate results obtained by the direct integral.

The numerical examples indicate that SDP-FLAM is specially suit for the non-near field Green's function, while DCIM can have high precision for the near field Green's function, and hence these two methods can supplement each other. According to refs. [34, 35] and our practices, $\rho = 0.05\lambda$ can be chosen as a dividing point for these two methods.

## 3 Conclusion

In this paper, based on local Taylor expansions on the complex plane, a method for fast locating all modes (FLAM) of spectral-domain Green's functions in a planar layered medium has been developed. SDP-FLAM, the combination of FLAM with SDP, can be used to accurately calculate the spatial-domain Green's functions in a layered medium. A prominent advantage of SDP-FLAM over NMSP-FAM[35], is its ability of ensuring its validity in any large range. The numerical examples presented in this paper have validated the correctness and effectiveness of SDP-FLAM, and indicated that SDP-FLAM is specially suit for the non-near field Green's function, while DCIM can have high precision for the near field Green's function, and hence these two methods can supplement each other very well. In our future work, SDP-FLAM method will be extended to more complicated structures and applied to the full wave analysis of microwave and millimeter wave LTCC passive integrated circuit.


1 Sommerfeld A. Propagation of waves in wireless telegraphy. Ann Phys, 1909, 28: 665–736

2 Kong J A. Electromagnetic Wave Theory. New York: Wiley, 1981

3 Chew W C. Waves and Fields in Inhomogeneous Media. New York: Van Nostrand, 1990

4 Lindell I V, Alanen E. Exact image theory for the Sommerfeld half space problem, Pt. 3: general formulation. IEEE Trans Antennas Propag, 1984, 32(10): 1027–1032

5 Fang D G, Yang J J, Delisle G Y. Discrete image theory for horizontal electric dipoles in a multilayered medium. IEE Proc H, 1988, 135(5): 297–303

6 Chow Y L, Yang J J, Fang D G, et al. A closed-form spatial Green's function for the thick microstrip substrate. IEEE Trans Microwave Theory Tech, 1991, 39: 588–592

7 Kipp R A, Chan C H. Complex image method for sources in bounded regions of multilayer structures. IEEE Trans Microwave Theory Tech, 1994, 42: 860–865

8 Dural G, Aksun M I. Closed-form Green's functions for general sources and stratified media. IEEE Trans Microwave Theory Tech, 1995, 43: 1545–1551

9 Hua Y, Sakar T K. Generalized pencil-of-function method for extracting poles of an EM system from its transient response. IEEE Trans Antennas Propagat, 1995, 37: 229–234

10 Aksun M I. A robust approach for the derivation of closed-form Greens functions. IEEE Trans Microwave Theory Tech, 1996, 44: 651–658

11 Kinayman N, Aksun M I. Efficient use of closed-form Green's functions for the analysis of planar geometries with vertical connections. IEEE Trans Microwave Theory Tech, 1997, 45:

593–603
12 Michalsky K A, Mosig J R. Multilayered media Greens functions in integral equation formulations. IEEE Trans Microwave Theory Tech, 1997, 45: 508–519
13 Cui T J, Chew W C. Fast evaluation of Sommerfeld integrals for EM scattering and radiation by three dimension buried objects. IEEE Trans Geosci Remote Sens, 1999, 37: 887–900
14 Ling F. Fast electromagnetic modeling of multilayer microstrip antennas and circuits. Ph. D. Thesis in Elect. Eng. Urbana-Champaign: Illinois Univ., 2000
15 Ling F, Jin J M. Discrete complex image method for Greens functions of general multilayer media. IEEE Trans Microw Guided Wave Lett, 2000, 10: 400–402
16 Ling F, Liu J, Jin J M. Efficient electromagnetic modeling of three-dimensional multilayer microstrip antennas and circuits. IEEE Trans Microwave Theory Tech, 2002, 50: 1628–1635
17 Ling F, Okhmatovski V, Song B, et al. Systematic extraction of static images from layered Media Green's function for accurate DCIM implementation. IEEE Antennas Wireless Propag Lett, 2007, 6: 215–218
18 Liu Y, Li L W, Yeo T S, et al. Application of DCIM to MPIE-MoM analysis of 3-D PEC objects in multilayered media. IEEE Trans Antennas Propag, 2002, 50: 157–162
19 Liu Y, Li L W, Yeo T S, et al. Calculation of spatial-domain Green functions for multilayered media using DCIM with automatic handling of surface wave poles. IEE Proc-Microw Antennas Propag, 2004, 151: 236–240
20 Liu E X, Li E P, Li L W. Electrical performance simulation of inhomogeneous multilayer LTCC structure by hybrid method. In: Electronics Packaging Technology Conference, 2004. 657–660
21 Bernal J, Medina F, Boix R R, et al. Fast full-wave analysis of multistrip transmission lines based on MPIE and complex image theory. IEEE Trans Microwave Theory Tech, 2000, 48(3): 445–452
22 Simsek E, Liu Q H, Wei B. Singularity subtraction for evaluation of Green's function for multilayer media. IEEE Trans Microwave Theory Tech, 2006, 54: 216–225
23 Simsek E, Liu J G, Liu Q H. A spectral integral method (SIM) for layered media. IEEE Trans Antennas Propag, 2006, 54: 1742–1749
24 Hsieh R C, Kuo J T. Evaluation of spatial Green's functions for microstrips: fast Hankel transform algorithm and complex image method. Electr Lett, 1998, 34(11): 1110–1111
25 Song B, Hong W. Extraction of the surface waves and waveguide modes of the Green's function in layered anisotropic dielectrics between two ground planes. Radio Sci, 2002, 37(4)
26 Kourkoulos V N, Cangellaris A C. Accurate approximation of Greens functions in planar stratified media in terms of a finite sum of spherical and cylindrical waves. IEEE Trans Antennas Propag, 2006, 54(5): 1568–1576
27 King R W P. The electromagnetic field of a horizontal electric dipole in the presence of a three-layered region. J Appl Phys, 1991, 69: 7985–7995
28 King R W P, Sandler S S. The electromagnetic field of a vertical electric dipole in the presence of a three-layered region. Radio Sci, 1994, 29: 97–113
29 Zhang H Q, Pan W Y. Electromagnetic field of a vertical electric dipole on a perfect conductor coated with a dielectric layer. Radio Sci, 2002, 37: 13-1–13-7
30 Li K, Lu Y. Electromagnetic field generated by a horizontal electric dipole near the surface of a planar perfect conductor coated with a uniaxial layer. IEEE Trans Antennas Propag, 2005, 53: 3191–3200
31 Teo S A, Chew S T, Leong M S. Error analysis of the discrete complex image method and pole extraction. IEEE Trans Microwave Theory Tech, 2003, 51: 406–413
32 Neve M J, Paknys R. A technique for approximating the location of surface- and leaky-wave poles for a lossy dielectric slab. IEEE Trans Antennas Propag, 2006, 54(1): 115–120
33 Tsang L, Ong C J, Wu B. Electromagnetic fields of Hertzian dipoles in thin-layered media. IEEE Antennas Wireless Propag Lett, 2006, 5: 537–540
34 Tsang L, Wu B. Electromagnetic fields of Hertzian dipoles in layered media of moderate thickness including the effects of all modes. IEEE Antennas Wireless Propag Lett, 2007, 6: 316–319
35 Wu B, Tsang L, Ong C J. Fast all modes (FAM) method combined with NMSP for evaluating spatial domain layered medium Green's functions of moderate thickness. Microw Opt Tech Lett, 2007, 49(12): 3112–3118
36 Marin M A, Barkeshli S, Pathak P H. On the location of proper and improper surface wave poles for the grounded dielectric slab. IEEE Trans Antennas Propag, 1990, 38: 570–573
37 Jackson D R, Oliner A A. A leaky-wave analysis of the high-gain printed antenna configuration. IEEE Trans Antenna Propag, 1988, 36: 905–910
38 Lang S. Complex Analysis. 4th ed. New York: Spriger-Verlag, 1999
39 Fang D G. Antenna Theory and Microstrip Antennas (in Chinese). Beijing: Science Press, 2006
40 Gander W, Gautschi W. Adaptive quadrature-revisited. BIT, 2000, 40: 84–101